STATS 200A Lecture 9

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Definition 9.1
σ-algebras $\mathcal{G}_1, \ldots, \mathcal{G}_n \subseteq \mathcal{F}$ are *independent* if $\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n \mathbb{P}(A_i)$ for all $A_i \in \mathcal{G}_i$. Random elements X_i are independent if the $\sigma(X_i)$ are. Infinite families: every finite subfamily.

Theorem 9.2
$X_1, \ldots, X_n$ are independent iff $\mathbb{P}(X_1, \ldots, X_n) = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}$

Example 9.3

$$\mathbb{E}\{h(X)g(Y)\} = \mathbb{E}\{h(x)\}\,\mathbb{E}\{g(Y)\}$$

Solution.

$$\begin{aligned}
\int h(X)g(Y)\,d\mathbb{P} &= \int h(x)g(y)\,\underbrace{d\mathbb{P}_{X,Y}(x,y)}_{\mathbb{P}_{X,Y}}(dx,dy) \\
&= \int \left(\int h(x)g(y)\,\mathbb{P}_X\,(dx)\right)\mathbb{P}_Y(dy) \\
&= \int g(y)\left[\int h(x)\,\mathbb{P}_X(dx)\right]\mathbb{P}_Y(dy) \\
&= \mathbb{E}(h(X))\mathbb{E}(g(Y))
\end{aligned}$$

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Definition 9.4 (Coupling)
Given margins $\mu_1, \ldots, \mu_n$, a *coupling* is any joint law with those marginals. The product law is the independent coupling.

Theorem 9.5
μ_n, probability measures on $\mathbb{R}^n$, consistet $\mu_{n+1}(B \times \mathbb{R}) = \mu_n(B)$ for every $B \in \mathcal{B}(\mathbb{R}^n)$. Then there is a unique probability measure on $(\mathbb{R}^{\mathbb{N}}, \mathcal{B}(\mathbb{R}^{\bigotimes \mathbb{N}}))$ with first n-coordinate laws μ_n.

Remark 9.6

$$\mathbb{E}\,|UV| \leq \sqrt{\mathbb{E}(U^2)\mathbb{E}(V^2)}$$

$$\langle x, y\rangle = \sum_{i=1} x_i y_i$$
$$|\langle x, y\rangle| \leq \|x\|_2 \, \|y\|_2$$
$$\|x\|_2 = \sqrt{\langle x, x\rangle} = \sqrt{\sum_i x_i^2}$$

Remark 9.7 (Best constant prediction)

$$\arg\min_{c\in\mathbb{R}} \mathbb{E}\,(X-c)^2 = \mathbb{E}(X), \qquad \text{minimum error } = \mathrm{Var}(X)$$

Proof. Expand $\mathbb{E}(X-c)^2 = \mathrm{Var}(X) + (\mathbb{E}(X)-c)^2$.

$$\mathbb{E}|left(X-m+m-c)^2 = \mathbb{E}\left((x-m)^2\right) + (m-c)^2 + \text{ cross term}$$

Geometrically: constants form a line on L^2. The residua $\mathrm{l}X - \mathbb{E}(X)$ is orthogonal to that line:

$$(X-\mathbb{E}(X), 1) = \mathbb{E}(X-\mathbb{E}(X)) = 0$$

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Remark 9.8 (The space of allowed predictions)

$$S_Y = \left\{g(Y) : g : \mathbb{R} \to \mathbb{R} \text{ Borel }, \mathbb{E}\left[g(Y)^2\right] < \infty\right\}.$$

Choose prediction that minimizes squared error:

$$\mathbb{E}\left[X \mid Y\right] := g^*(Y) = \arg\min_{Z\in S_Y} \mathbb{E}\,(X-Z)^2\,.$$

S_Y is a closed lienar subspace of L^2, so the best predicion is the orthogonal projection of X onto it. If we only know if our die is odd or even, then we will predict 3 if odd and 4 if even.

Remark 9.9 (Best prediction at finite resolution)

$$M = g^*(Y) = \sum_{i:\mathbb{P}(A_i)>0} m_i \mathbf{1}_{A_i}, \qquad g^*(y_i)m_i = \frac{\mathbb{E}[X\mathbf{1}_{A_i}]}{\mathbb{P}(A_i)} = \frac{\int_{A_i} X(\omega)\, d\mathbb{P}(\omega)}{\int_{A_i} d\mathbb{P}(\omega)}$$

Theorem 9.10 (Doob-Dynkin)

Z is $\sigma(Y)$-measurable iff $Z = g(Y)$ for some Borel g.

$$\boxed{S_Y = L^2(\sigma(Y))}$$

More generally, let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra. $L^2(\mathcal{G})$ consists of all square-integrable $\mathcal{G}$-measurable variables.

Definition 9.11 (Conditioning on Information)

For $X \in L^2$,

$$\mathbb{E}[X \mid \mathcal{G}] = \underset{Z \in L^2(\mathcal{G})}{\arg\min} \mathbb{E}(X - Z)^2$$

In particular, $\mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \sigma(Y)]$.

Definition 9.12 (Orthogonality)

$$\mathbb{E}[RW] = 0 \qquad \forall W \in L^2(\mathcal{G})$$

Since $M - Z \in L^2(\mathcal{G})$, Pythagoras gives

$$\underbrace{\mathbb{E}(X - Z)^2}_{\text{error of any prediction}} = \underbrace{\mathbb{E}R^2}_{\text{minimum error}} + \underbrace{\mathbb{E}(M - Z)^2}_{\text{extra error}}$$